

# Lab-1 Assignment

---

## Procedure

### Accept the assignment

Submit all your code to your repository under Lab 1 folder by the due date time.

### Phase 1

- Blinky! (The hello world of embedded)
- Start the STM32CubeIDE software and create a new STM32 project.
- Search and Select our Nucleo-64 board.
- Initiate all peripherals with their default mode.
- Use the graphical tool to ensure that PA5 is configured to connect to the Green LED on the board.
- Save the project and generate all the setup code.
- Look at the HAL documentation (20.2.4) to toggle this light and delay (6.1.3) for 1 second. Make use of the HAL\_Delay function.
- Make sure you check in your code and then push to your remote repository. Use appropriate commit messages.
- 

### Phase 2

Now that you can control the light you will get rid of this blocking delay function and use timers.

- Look at the Clock tree (2.1 - Figure 1 and Figure 2 of the STM32F103RB data sheet) and notice that the clock source for Timer 4 is the APB1 clock.
- Open the \*.ioc file that graphically allows you to configure the H/W for your board.
- Under the Pinout and Configuration Tab open the Timers section for the Categories
- Enable Timer 4 by selecting the Internal Clock. Notice that you can configure the parameters for the clock below in the Parameter Settings such as prescaler etc.
- Now Open the Clock Configuration tab (Up above beside the Pinout & Configuration tab). Notice the clock speed for APB1.
- Look at the code generated in the configuration to understand how to update the prescaler and timer period.
- Consult the HAL documentation to get the timer started and to read it's current value. HAL\_TIM\_Base\_Start() and the macro HAL\_TIM\_GET\_COUNTER(). You need to pass in the handler for this timer which is given to you when you generate the code.
- Based on this, set the pre-scaler to any desired number and get the light to blink every 1 second.

## Submission

Make sure you have committed your latest solution and pushed it to your repository for grading.